Computer Vision - Fall 2025 Project Proposal

Advanced Lane Detection for Autonomous Driving: A Comparative Analysis of Classical and Deep Learning Approaches

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1. Introduction

Advanced Driver-Assistance Systems (ADAS) and autonomous vehicle technologies represent one of the most active research areas in modern engineering. The safe and effective operation of these systems is critically dependent on an accurate perception of the vehicle's environment. Lane detection is one of the most fundamental computer vision tasks required for this, enabling the vehicle to localize itself on the road and plan its path.

On well-lit, clearly-marked highways, classical computer vision techniques (such as edge detection and line fitting) can successfully detect lanes. However, "real-world" conditions are far from this ideal scenario. Shadows, adverse weather conditions like rain, poor nighttime illumination, and reflections on the road surface can easily cause these classical methods to fail.

We will develop and compare two distinct lane detection systems:

1. **Classical Approach:** A traditional pipeline, specifically using Canny Edge Detection and the Hough Transform.
2. **Deep Learning Approach:** A modern system using a U-Net architecture for semantic segmentation to distinguish lanes at a pixel level.

We will quantitatively (FPS) and qualitatively (visual) analyze the performance of these two approaches, especially under challenging test scenarios (nighttime and rainy conditions).

2. Problem Statement

Problem: The core problem is to develop a system capable of reliably detecting lane markings not only in ideal daylight conditions, but also in low-light (night) and reflective (rainy) environments. The project will then compare this system's performance against a classical method.

Dataset to be used:

Our project will use two distinct datasets, clearly separating the training and testing phases:

1. **Training Dataset (for U-Net): BDD100K (Berkeley DeepDrive)**
    - To train our deep learning model (U-Net), we require tens of thousands of varied images. BDD100K is a massive dataset with over 100,000 videos, encompassing diverse weather

(rain, etc.) and time-of-day (night, etc.) conditions. Most importantly, it includes the pixel-level segmentation labels for "lane markings" and "drivable areas" that we need for training our model.

2. **Test and Comparison Dataset: Driving video for lane detection (Various weather)**
    - This 1.2GB set of 60 videos is perfect for testing our models on unseen data. It contains the "Normal Day," "Normal Night," and "Rainy Day" scenarios that are critical for our project's main comparative goal.

Expected Results and Evaluation:

- **Expectation:** We expect the classical Hough method to perform well on the "Normal Day" videos but fail or produce erroneous detections in the "Rainy Day" (due to reflections) and "Normal Night" (due to low contrast) videos. We expect the U-Net model, trained on BDD100K, to remain more robust in these challenging conditions, correctly segmenting the lanes.
- **Evaluation Metrics:**
    1. **Qualitative (Visual) Evaluation:** The main output will be side-by-side comparison videos showing both methods running on the same test footage. In these videos, the detected lane area will be visualized as a green polygon, similar to the example screenshot.
    2. **Quantitative (Numerical) Evaluation:** We will compare the processing speed (FPS) of the two approaches to analyze the trade-off between accuracy and computational cost.

3. Technical Approach

Our project will consist of two main pipelines developed using Python, with OpenCV for the classical method and PyTorch/TensorFlow for the deep learning method.

Approach 1: Classical (Hough Transform) Pipeline

This system will implement traditional CV techniques from the course in a step-by-step manner:

1. **Image Preprocessing:** Each frame from the test video will be converted to grayscale and a Gaussian Blur will be applied to reduce noise.
2. **Edge Detection:** The Canny Edge Detection algorithm will be used to find sharp intensity changes in the image.
3. **Region of Interest (ROI) Masking:** A mask-like trapezoid shape will be applied to focus only on the bottom half of the image where the road is expected, filtering out edges from irrelevant areas like the sky or trees.
4. **Line Detection:** The Hough Transform will be applied to the masked edge image to detect straight lines corresponding to the lane markings.
5. **Line Averaging:** The multiple line segments found by Hough will be grouped (based on slope, into left/right lanes) and averaged to produce a single, stable left and right lane line.

Approach 2: Deep Learning (U-Net) Pipeline

This system will implement the deep learning techniques specified in the project document:

1. **Model Architecture:** We will use a U-Net architecture for the semantic segmentation task. U-Net's encoder-decoder structure is ideal for capturing both contextual information and precise localization
2. **Training:** Our U-Net model will be trained on the thousands of labeled images from the **BDD100K** dataset. The model will learn to output a probability mask, classifying each pixel as either background or lane marking.
3. **Inference:** The trained model will take raw frames from the test videos as input and output a segmentation mask identifying the location of the lanes.

Comparison and Visualization:

The lane information from both systems (Hough lines and U-Net mask) will be visualized on the input frame. Side-by-side videos will be created to demonstrate the performance gap in challenging scenarios (rain/night).

4. References

1. BDD100K: A Large-scale Diverse Driving Video Database. (Kaggle Dataset)
   - https://www.kaggle.com/datasets/marquis03/bdd100k
2. Driving video for lane detection (Various weather). (Kaggle Dataset)
   - https://www.kaggle.com/datasets/ashikadnan/driving-video-for-lane-detection-various-weather
3. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *arXiv*.
4. Yu, F., et al. (2020). BDD100K: A Large-scale Diverse Driving Video Database. *arXiv*.
5. Hough, P. V. C. (1962). Method and means for recognizing complex patterns. *U.S. Patent 3,069,654*.